

Javascript

Es6+, encapsulation

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Encapsulation</title>
  </head>
  <body>
    <script>
      // encapsulation => bind data with method

      //   class Employee {
      //     #salary = 10000;

      //     constructor(name, salary) {
      //       this.name = name;
      //       this.#salary = salary;
      //     }
      //     display() {
      //       console.log(this.#salary);
      //     }

      //     withdraw(amount) {
      //       if (this.#salary >= amount) {
      //         this.#salary -= amount;

      //         console.log("Balance amount", this.#salary);
      //       } else {
      //         console.log("Insufficient balance");
      //       }
      //     }
      //   }

      //   let obj = new Employee("Arun", 30000);
      //   console.log(obj.salary);

      //   obj.withdraw(50000);

      //   obj.display();

      //   Es6+

      //   // spread operator , rest parameter
      //   const arr = [1, 2];
```

```

//  const arr2 = [...arr, 3, 4];
//  console.log(arr2);

//  const objj2 = { academy: "IT insittue", branch: "TVM" };

//  const objj3 = { ...objj, ...objj2 };

//  function insittue(...objj2) {
//    console.log(...objj2);
//  }

//  insittue("Arun", 23, "iit");

//  console.log(objj);

//  destructing
const objj = { name: "FIIT", course: "Python full stack" };

console.log(objj.name);
console.log(objj.course);

const { name, course } = objj;

console.log(name);
console.log(course);

//template literal

</script>
<script src="./export.js"></script>
</body>
</html>

```

Theory Notes

1. Encapsulation

- **Definition:** Binding data and methods together, restricting direct access to data.
- Achieved in JavaScript using:
 - Private fields (#fieldName) → accessible only inside the class.
 - Getter/Setter methods → controlled access to private data.
- Benefits: Security, controlled modification, data hiding.

2. ES6 Features

- **Spread operator (...)** → expands arrays/objects.

```
const arr1 = [1,2];
```

```
const arr2 = [...arr1, 3,4]; // [1,2,3,4]
```

- **Rest parameter (...args)** → collects multiple arguments into an array.

```
function sum(...nums){ return nums.reduce((a,b)=>a+b); }
```

- **Destructuring** → extract values from objects/arrays easily.

```
const obj = {name:"Arun", course:"JS"};
```

```
const {name, course} = obj;
```

3. Template Literals

- Defined using backticks ``.
- Allow string interpolation:

```
let name = "Arun";
```

```
console.log(`Hello, ${name}!`);
```

Coding Round Questions

1. Create a class Employee with a private field #salary. Add methods to display and withdraw salary. Demonstrate encapsulation.
2. Use the spread operator to merge two arrays [1,2] and [3,4].
3. Write a function using rest parameters that accepts any number of arguments and prints them.
4. Create an object {academy:"FIIT", branch:"TVM"} and merge it with another object using spread operator.
5. Demonstrate object destructuring by extracting name and course from {name:"Priya", course:"Python"}.
6. Write a program using template literals to print:

Hello, my name is Arun and I study Python full stack.